The minimal dispersion in the unit cube.

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based on joint works with **A. Arman** and **G. Livshyts**.

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Equivalently: Given integers $n, d \geq 1$ what is the supremum over all $\varepsilon > 0$ such that for any n points in the unit d-dimensional cube $[0, 1]^d$ there exists an axis-parallel box of volume ε containing none of these points?

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Our goal is to understand the behaviour of $N(\varepsilon, d)$ as $\varepsilon \to 0$ and $d \to \infty$.

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Consider the set of all axis parallel boxes contained in the cube $[0,1]^d$,

$$\mathcal{R}_d := \left\{ \prod_{i=1}^{d} I_i \;\mid\; I_i = [a_i, b_i) \subset [0,1] \right\}.$$

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Their result also implies that for every (fixed) d there exists $\lim\limits_{\varepsilon \to 0} \varepsilon N(\varepsilon, d) = c_d$.

Thus, when the dimension d is fixed and $\varepsilon \to 0$ the problem is essentially solved:

$$N(\varepsilon, d) \approx \frac{c_d'}{\varepsilon}.$$

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The first bound showing that the minimal dispersion grows with the dimension was obtained by Aistleitner–Hinrichs–Rudolf (**17**):

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Thus $\forall \varepsilon \in (0, 1/4) \; :$ $$\frac{\log_2 d}{8\varepsilon} \leq N(\varepsilon, d) \leq \frac{Cd^2 \log d}{\varepsilon}.$$

and $$\frac{d}{e} \leq c_d := \lim_{\varepsilon \to 0} \varepsilon N(\varepsilon, d) \leq Cd^2 \log d.$$

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Very recent intermediate lower bounds by Trödler–Volec–Vybíral (**25**), in particular

$$\forall \varepsilon < c/\sqrt{d} : \qquad N(\varepsilon, d) \geq C\sqrt{d}/\varepsilon.$$

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A natural conjecture: $N(\varepsilon, d) \approx \frac{d}{\varepsilon}$.

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Trödler–Volec–Vybíral (**24**): $$C_\varepsilon \ge \frac{C}{\varepsilon^2 \log(1/\varepsilon)}, \quad \textit{whenever} \quad \frac{1}{4} \ge \varepsilon \ge \frac{1}{4\sqrt{d}}$$

The S.–U.–V. proof is also based on a random choice of points, but they use uniform distribution on a certain lattice, gaining in the case of relatively large ε.

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Remarks on very large ε.

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Kurt MacKay (**22**): $N(\varepsilon, d) \leq \dfrac{C}{\sqrt{\varepsilon - 1/4}}$ for $\varepsilon > 1/4$.

Upper bounds: summary

$$N(\varepsilon, d) \leq \begin{cases} \dfrac{C\ \ln d}{\varepsilon^2}\ \ln\left(\dfrac{1}{\varepsilon}\right), & \text{if} \quad \varepsilon \geq \dfrac{\ln^2 d}{d}, \\ \dfrac{C\, d}{\varepsilon}\ \ln\left(\dfrac{1}{\varepsilon}\right), & \text{if} \quad \dfrac{\ln^2 d}{d} \geq \varepsilon \geq \exp(-Cd\ln d), \\ \dfrac{C\, d^2 \ln d}{\varepsilon}, & \text{if} \quad \varepsilon \leq \exp(-Cd\ln d). \end{cases}$$

New bounds.

Theorem (Arman–L. 24)

Let $d \geq 2$ *and* $\varepsilon \leq 1/2$. *Then*
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Remarks. 1. This result improves the previous results L. (**21**) and Livshyts–L. (**22**):

$$N(\varepsilon, d) \leq 4e\, \frac{2d \ln\ln(8/\varepsilon) + \ln(1/\varepsilon)}{\varepsilon},$$

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$$\frac{d + c\ln(1/\varepsilon)}{2\varepsilon} \leq N_{ran}(\varepsilon, d).$$

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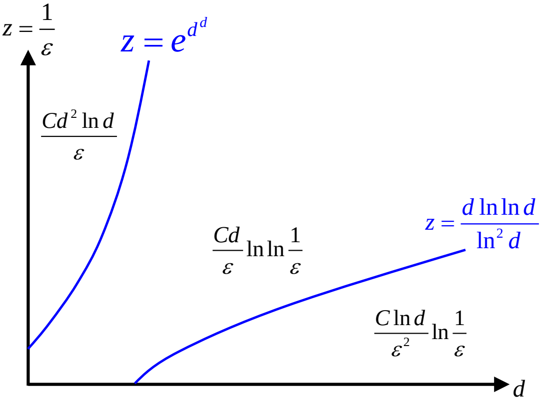
3. Recall Rudolf and Bukh–Chao bounds

$$N(\varepsilon, d) \leq \frac{Cd}{\varepsilon} \ln\left(\frac{1}{\varepsilon}\right) \qquad \text{and} \qquad N(\varepsilon, d) \leq \frac{Cd^2 \log d}{\varepsilon}.$$

The current state of the art (upper bounds).

$$N(\varepsilon, d) \leq \begin{cases} \frac{C\ \ln d}{\varepsilon^2}\ \ln\left(\frac{1}{\varepsilon}\right) & \text{if } \varepsilon \geq \frac{\ln^2 d}{d \ln\ln(2d)}, \\ \frac{C\, d}{\varepsilon}\ \ln\ln\left(\frac{1}{\varepsilon}\right) & \text{if } \frac{\ln^2 d}{d\ln\ln(2d)} \geq \varepsilon \geq \exp\left(-d^d\right), \\ \frac{C\, d^2\ \ln d}{\varepsilon} & \text{if } \varepsilon \leq \exp\left(-d^d\right), \end{cases}$$



Some ideas of the proof (for random choice of points).

Consider a set P of random points independently and uniformly distributed in the cube. To show that (with good probability) every box of volume ε contains at least one point from P, we need to construct a set $\mathcal{N}$ of "test boxes." It should satisfy

each rectangle in $\mathcal{N}$ *contains a point from* $P \qquad \Longrightarrow$

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Individual bounds are simple — volume computations
(need that volumes of boxes in $\mathcal{N}$ are not to small — comparable to ε).

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(need that volumes of boxes in $\mathcal{N}$ are not to small — comparable to ε).

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Consider a set P of random points independently and uniformly distributed in the cube. To show that (with good probability) every box of volume ε contains at least one point from P, we need to construct a set $\mathcal{N}$ of "test boxes." It should satisfy

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Denote
$$\mathcal{B}_{\varepsilon,d} := \Big\{ B \in \mathcal{R}_d \;\mid\; |B| \geq \varepsilon \Big\}.$$

Definition. *We say that* $\mathcal{N} \subset \mathcal{R}_d$ *is a* δ*-approximation for* $\mathcal{B}_{\varepsilon,d}$ *if*
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Lemma

Let $d \geq 1$ *and* $\varepsilon, \delta \in (0,1)$. *Let* $\mathcal{N}$ *be a* δ*-approximation for* $\mathcal{B}_{\varepsilon,d}$ *with* $\delta|\mathcal{N}| \geq e$. *Then*

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Remark. A variant of a lemma proved by Rudolf shows that with probability at least $1 - 1/|\mathcal{N}|$ a random choice of $N = \left\lfloor \frac{3\ln|\mathcal{N}|}{\delta} \right\rfloor$ points satisfies the desire property.

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Remark. The random choice corresponds to the event $\ell = 0$.

Construction of a net. Step I.

We first construct an approximation for the set of so-called anchored boxes,

$$\mathcal{R}_d^0 := \Big\{ \prod_{i=1}^{d} I_i \;\mid\; I_i = [0, b_i) \subset [0,1] \Big\}.$$

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For $\beta \geq 1$ consider the sets $\quad \mathcal{A}_d(\varepsilon^\beta) = \left\{ b = \{b_i\}_{i=1}^d \in [0,1]^d \;\middle|\; \prod_{i=1}^{d} b_i = \varepsilon^\beta \right\}.$

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For some $\gamma > 0$ we want to construct a set $\mathcal{N}_0 \subset \mathcal{A}_d(\varepsilon^{1+\gamma})$ such that

$$\forall\, b = \{b_i\}_{i=1}^d \in \mathcal{A}_d(\varepsilon) \quad \exists a = \{a_i\}_{i=1}^d \in \mathcal{N}_0 \;:\quad a_i \leq b_i \quad \text{for each } i.$$

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Thus, our problem reduces to approximating points from

$$S_{d-1} := \Big\{x \,\Big|\, \sum_{i=1}^d x_i = 1,\ x_i \geq 0\Big\} \quad \text{by points from} \quad \Big\{y \,\Big|\, \sum_{i=1}^d y_i = 1+\gamma,\ y_i \geq 0\Big\}.$$

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This is turn reduces to covering the regular simplex S_{d-1} by shifts of $(-\gamma S_{d-1})$.
Such bounds are well-known.

Construction of a net. Step II.

With each $B = \prod_{i=1}^{d}[0, b_i) \in \mathcal{N}_0$ we associate the following set of points,

$$(*) \qquad \mathcal{L}_B := \left\{ z = \{k_i b_i/d\}_{i=1}^{d} \mid \forall i \le d : \; k_i \in \mathbb{Z}, \; 1 \le k_i \le 1 - d + d/b_i \right\}.$$

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Then

$$|\mathcal{L}_B| \le \prod_{i=1}^{d} \left(1 - d + \frac{d}{b_i}\right) \le \ldots \le \frac{(\ln(e/\delta_0))^d}{\delta_0}, \qquad \text{where } \ \delta_0 = \prod_{i=1}^{d} b_i = \varepsilon^{1+\gamma}.$$

Finally we take

$$\mathcal{N} := \bigcup_{B \in \mathcal{N}_0} \bigcup_{z \in \mathcal{L}_B} (z + (1 - 1/d)B), \qquad |\mathcal{N}| \le |\mathcal{N}_0| \cdot |\mathcal{L}_B|.$$

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Remark. In the case of periodic boxes we have to ask $k_i \le 1 + d/b_i$ in $(*)$, so

$$|\mathcal{L}_B| = \prod_{i=1}^{d} \left(1 + \frac{d}{b_i}\right) \le \prod_{i=1}^{d} \frac{2d}{b_i} = \frac{(2d)^d}{\delta_0}.$$

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Theorem (L. 21, relatively large ε)

Let $d \geq 2$ *and* $\frac{\ln d}{d} \leq \varepsilon \leq 1/2$. *Then*

$$N(\varepsilon, d) \leq \frac{C \, \ln d}{\varepsilon^2} \, \ln\left(\frac{1}{\varepsilon}\right).$$

Construction of a net, the case of large ε.

We adjust the uniform distribution on the cube by sending each random point

$$X = (x_1, \dots, x_d) \longmapsto Y = \varphi(X) = (\varphi(x_1), \dots, \varphi(x_d)),$$

where $\varphi : [0,1] \to [\varepsilon, 1-\varepsilon]$ is given by
$$\varphi(t) = \begin{cases} \varepsilon & \text{if } 0 \le t < \varepsilon, \\ t & \text{if } \varepsilon \le t \le 1-\varepsilon, \\ 1-\varepsilon & \text{if } 1-\varepsilon < t \le 1. \end{cases}$$

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Next let $$B = \prod_{i=1}^{d} [a_i, b_i) \subset [0,1]^d, \qquad \ell_i := b_i - a_i, \qquad |B| = \prod_{i=1}^{d} \ell_i \ge \varepsilon.$$

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Assume first that $\ell_1 \le \ell_2 \le ... \le \ell_n$. Since $\prod_{i=1}^{d} \ell_i \ge \varepsilon$, for $j \ge m = \frac{\ln(1/\varepsilon)}{\varepsilon}$,

$$\ell_j \ge \left(\prod_{i=1}^{j} \ell_i \right)^{1/j} \ge \varepsilon^{1/j} > 1 - \frac{\ln(1/\varepsilon)}{j} \ge 1 - \varepsilon.$$

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Thus, we don't need to approximate the segments with $j \ge m$. This essentially reduces the dimension to $m = \dfrac{\ln(1/\varepsilon)}{\varepsilon}$.

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$$|J| = m \quad \text{and} \quad \ell_i \geq 1 - \varepsilon \text{ whenever } i \notin J.$$

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Then we approximate B by

$$\prod_{i \in J} [a_i', b_i') \times \prod_{i \notin J} [0, 1],$$

where $\prod_{i \in J} [a_i', b_i')$ approximates $\prod_{i \in J} [a_i, b_i)$ as above.

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Then we have the bound for approximation in $\mathbb{R}^m$ multiplied by $\binom{d}{m}$.

Dispersion on the torus.

We consider periodic axis parallel boxes, that is, boxes of the form

$$\prod_{i=1}^{d} I_i(a_i, b_i), \quad a_i, b_i \in [0,1],$$

where

$$I_i(a,b) := \begin{cases} (a_i, b_i), & \text{whenever } 0 \le a_i < b_i \le 1, \\ [0,1] \setminus [b_i, a_i], & \text{whenever } 0 \le b_i < a_i \le 1. \end{cases}$$

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$$I_i(a,b) := \begin{cases} (a_i, b_i), & \text{whenever } 0 \le a_i < b_i \le 1, \\ [0,1] \setminus [b_i, a_i], & \text{whenever } 0 \le b_i < a_i \le 1. \end{cases}$$

Combining recent results of M. Ullrich (**18**) and Rudolf (**18**)

$$\frac{d}{\varepsilon} \le \widetilde{N}(\varepsilon, d) \le \frac{8d}{\varepsilon} \left(\ln d + \ln\left(\frac{8}{\varepsilon}\right)\right).$$

Dispersion on the torus.

We consider periodic axis parallel boxes, that is, boxes of the form

$$\prod_{i=1}^{d} I_i(a_i, b_i), \quad a_i, b_i \in [0,1],$$

where

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Note that the VC dimension of the set of periodic axis parallel boxes is of the order $d \ln d$ (Gillibert-Lachmann-Müllner), therefore the BEHW result leads to

$$\widetilde{N}(\varepsilon, d) \le \frac{8d \ln d}{\varepsilon} \ln\left(\frac{8}{\varepsilon}\right)$$

— worse than the Rudolf bound.

Dispersion on the torus.

$$\frac{d}{\varepsilon} \le \widetilde{N}(\varepsilon, d) \le \frac{8d\ln d}{\varepsilon} + \frac{8d\ln\left(\frac{8}{\varepsilon}\right)}{\varepsilon}$$

We improve the Rudolf upper bound in the case $\varepsilon \le 1/d^C$.

Theorem (Arman–L. 24, bounds in the periodic case)

Let $d \ge 2$ *and* $\varepsilon \in (0, 1/2]$. *Then*

$$\widetilde{N}(\varepsilon, d) \le \frac{8ed\ln(2d)}{\varepsilon} + \frac{8ed\ln\ln(e/\varepsilon)}{\varepsilon}.$$

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Remark. For the random choice of points we can get

$$\widetilde{N}(\varepsilon, d) \le \frac{16ed \ln d}{\varepsilon} + \frac{8e \ln(1/\varepsilon)}{\varepsilon}.$$

(from the work with Livshyts (**22**)).